



Photo by Jason Masters



Co-funded by
the European Union

Project number: 2023-1-
FR01-KA220-SCH-
000151881

LET'S CODE ART



**Subject:**

Art & design

Content:

Lego press printing, crafting

**Computational practices:
choose what fits**

- Decomposing
- **Pattern recognition**
- **Abstraction**
- Algorithms
- Problem solving

Unplugged/plugged:

**Preparation time:**

30 min

**Age:**

6-10 years

**Duration:**

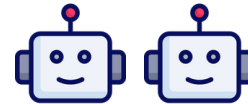
120 min

**Location:**

Classroom, art room

**Group distribution:**

in groups of two

**Difficulty****After this lesson, the children will be able to...**

- execute a sequence of steps correctly to achieve a desired outcome.
- understand and apply the basics of binary code using LEGO Dots.
- recognize patterns in programming cards and translate them into binary code.
- collaborate to decode programming instructions.
- express creativity by designing a LEGO Dots artwork.
- follow multi-step instructions independently.
- reflect on and appreciate their own and others' artistic creations.



Materials needed: for 20 kids:

- 1x Box to hold the following materials
 - 7x Paint rollers
 - Paint in various colors (posterpaint)
 - 1x Paper with QR code
 - 1x Stack of blank print paper
- 6x Combination locks (with a 4-digit code)
- 6x Symbol cards
- 24x Programming cards
- 24x Blank grid cards or checkered paper
- 6x Iron chains
- 1x LEGO Dots set
- 12x LEGO baseplates
- 3x Tape

Extra materials

- 12x Scissors
- 1x Box of markers/colored pencils
- 24x Pencils

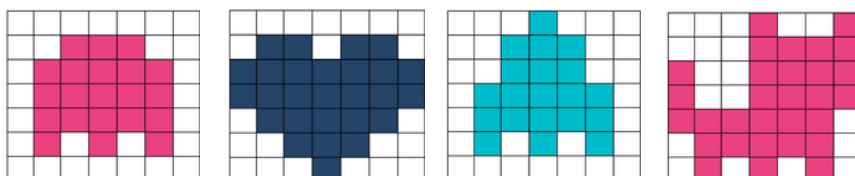
Helpful links:

Lego dot's printing: <https://www.youtube.com/watch?v=ebdl15nteck&t=16s>

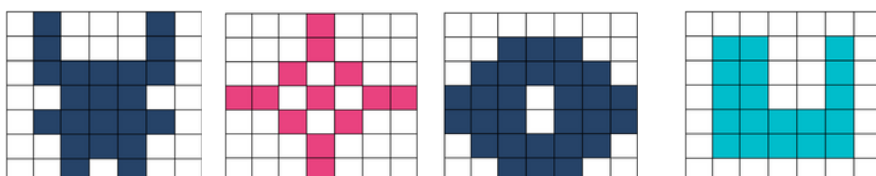
Information about binary code: <https://www.youtube.com/watch?v=USCBCmwMCDA>

locks box numbers:

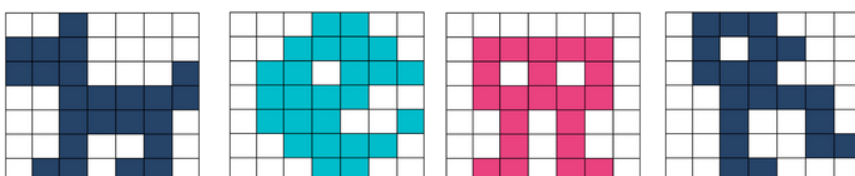
code lock A:5431



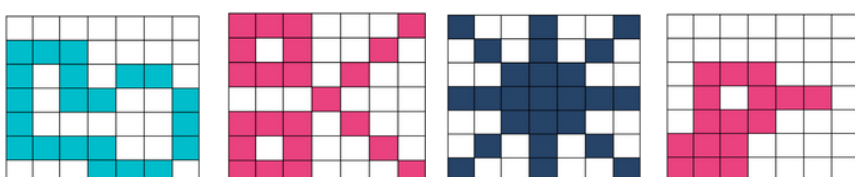
code lock B:8910



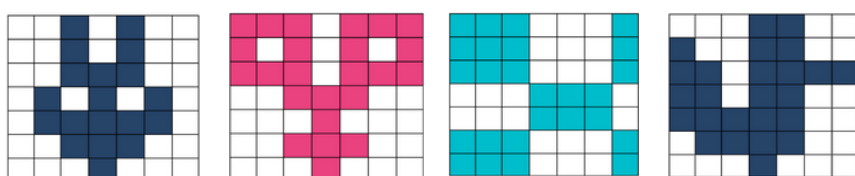
code lock C:6154



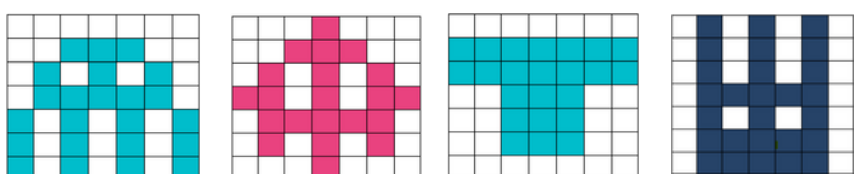
code lock D:3988



code lock E:4127



code lock F:9133



Preparation and teacher notes

Place the following items in the box:

- Paint (different colors)
- Paint roller
- Paper QR code
- Inspirational pictures

Then lock the box with the 6 number locks, use the iron chains for this if necessary.

Place this box in the middle of the classroom on a table and around the box place the programming cards.

Possible introduction: choose a story that features a treasure chest to introduce this chest. Or place a letter on the chest with "crack the code, if you dare."

Extra information: Binary is a number system that only uses two digits: 1 and 0. All information that is processed by a computer is in the form of a sequence of 1s and 0s. Therefore, all data that we want a computer to process needs to be converted into binary.

Secret box

Gather the children in a circle around the closed box.

Ask the following questions:

- What do you see?
- How can we open the box?



Activity instructions

Discuss the course of the activity with the children. Each combination lock has a piece of paper attached to it with four symbols, but no numbers. When looking at the programming cards placed around the box, we can see that they do contain numbers.

7							
2	1	2					
1	5	1					
1	5	1					
1	5	1					
1	1	1	1	1	1	1	1
7							

5

To determine which number belongs to which lock, the children must first decode the programming cards.

They do this by executing the program on the programming cards using LEGO Dots.

If they do it correctly, they will create a symbol that matches one found on a lock attached to the box. Here, we use binary code. All information processed by a computer consists of 1s and 0s—or in our case, a colored tile or an empty tile. In this way, we work just like a computer.

Important: Ensure that the number is entered in the correct position. For example, if the symbol appears second in the sequence, the number must be entered in the second position on the lock. If needed, complete one task together as a class.

Preparation and teacher notes

For more info on the Lego dot printing technique scan the QR code.



Tips for printing:

- Solid shapes require more pressure to print than lines. Standing up while pushing down during printing can help you get more leverage for better prints.
- Avoid printing on textured paper—smooth paper gives the best results.
- Before putting pieces away after printing, clean them off with a piece of scrap paper or a baby wipe to remove ink residue.

Activity:

The children form pairs. Each pair receives a LEGO baseplate and LEGO Dots. They choose two programming cards and solve them.

Next, they enter the correct number in the right position on the corresponding lock.

The pairs work together to unlock the box. Once all programming cards have been solved, they can use the information to open the chest.

Opening the box:

Gather the children around the box to open the final lock together.

Then, reveal the contents of the box to the children:

- Paint (various colors)
- Paint roller
- Paper with a QR code
- Inspiration photos

Scan the QR code and watch the video together as a class. In this video, they explain how to create a print artwork using LEGO Dots.

1. The children can now create their own Dots artwork on a LEGO baseplate. They can choose to work in pairs or individually.
2. Next, they roll paint onto the Dots blocks using the paint roller.
3. Finally, they place a piece of paper over the blocks and press it down firmly—making sure the paper doesn't shift.
4. Carefully lift the paper off the blocks and admire the printed artwork!

In the attachment, you will find several photos for inspiration. If needed, you can show these to the children.



**Scan me and
get started!**



Instagram: [Stirchley_printworks](#)



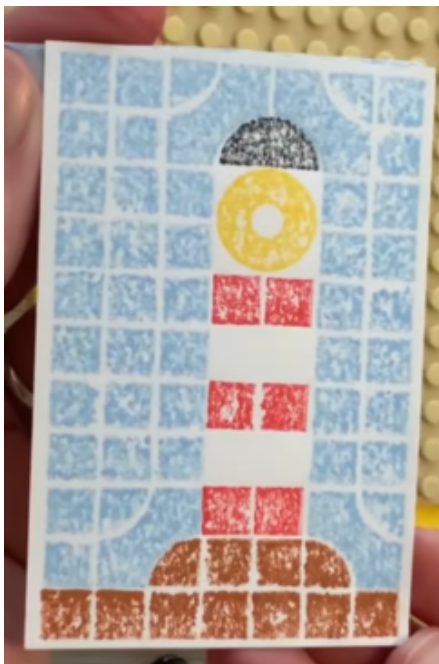
Instagram: provisionalpress

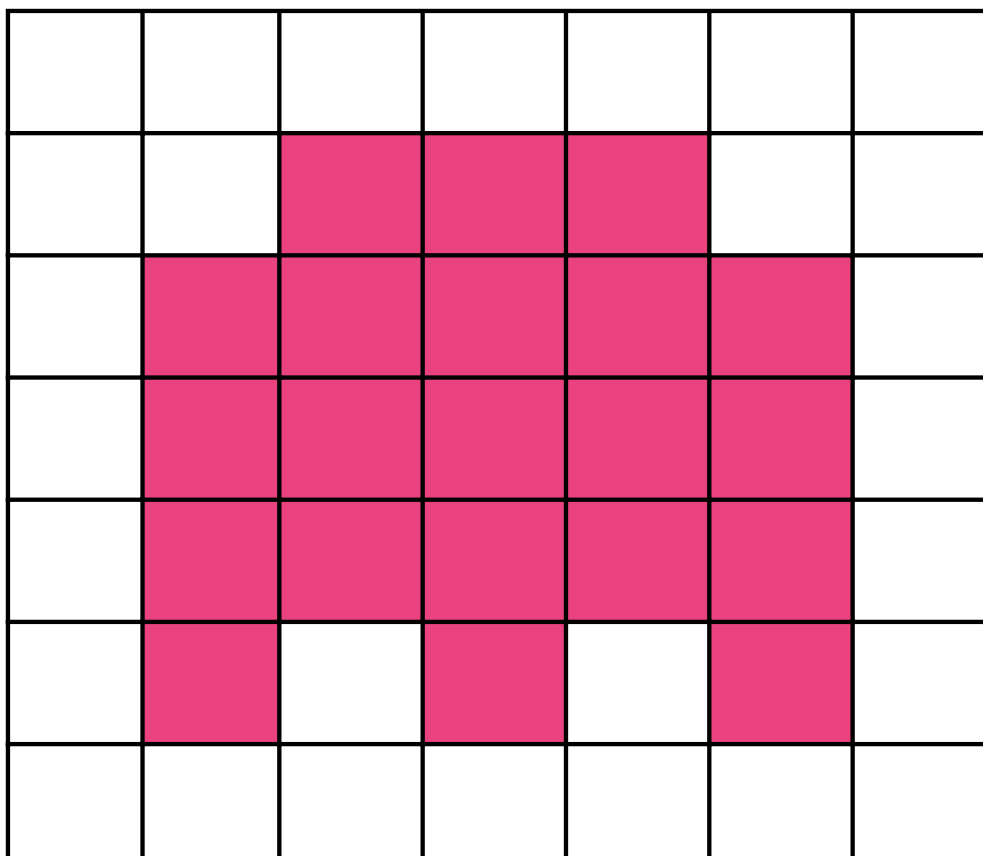


Instagram: Honeycomb.artistry



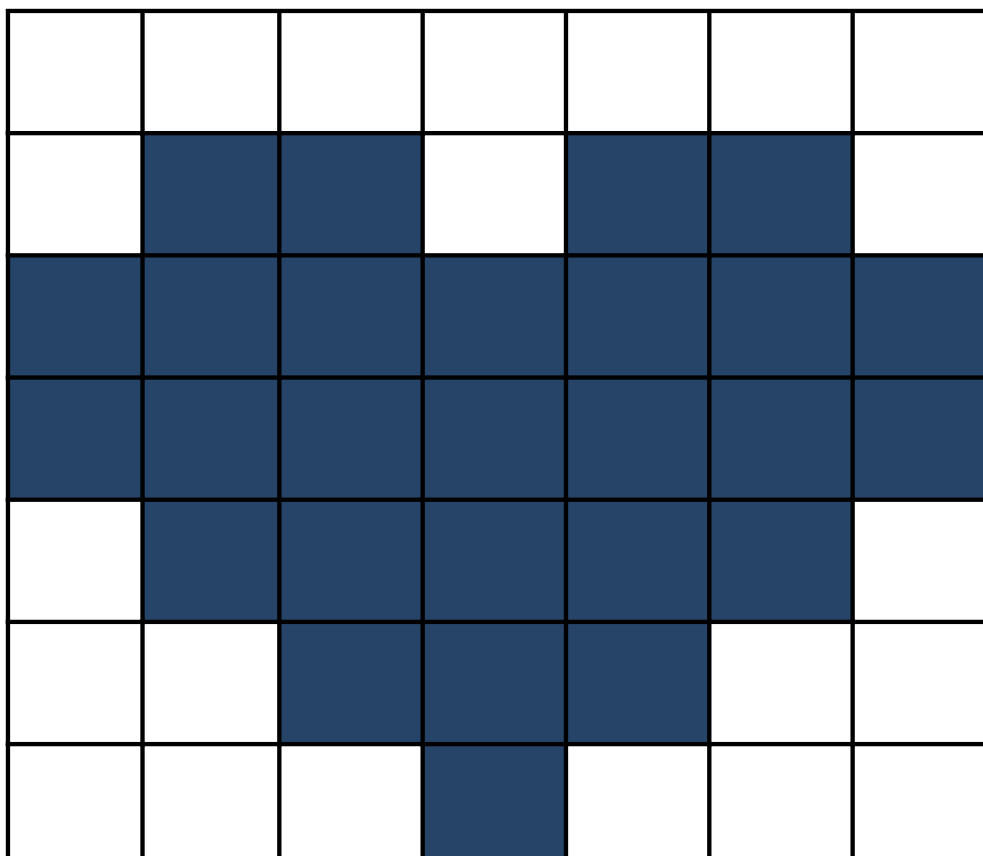
Instagram: Honeycomb.artistry





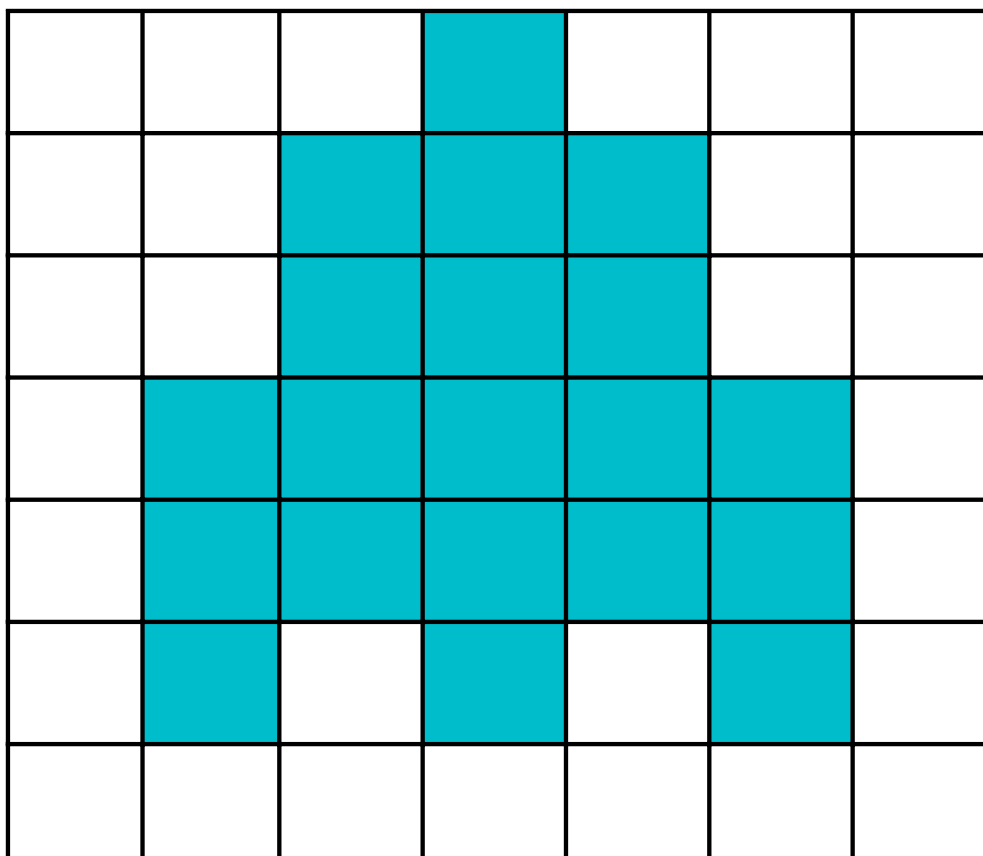
7						
2	3	2				
1	5	1				
1	5	1				
1	5	1				
1	1	1	1	1	1	1
7						

5



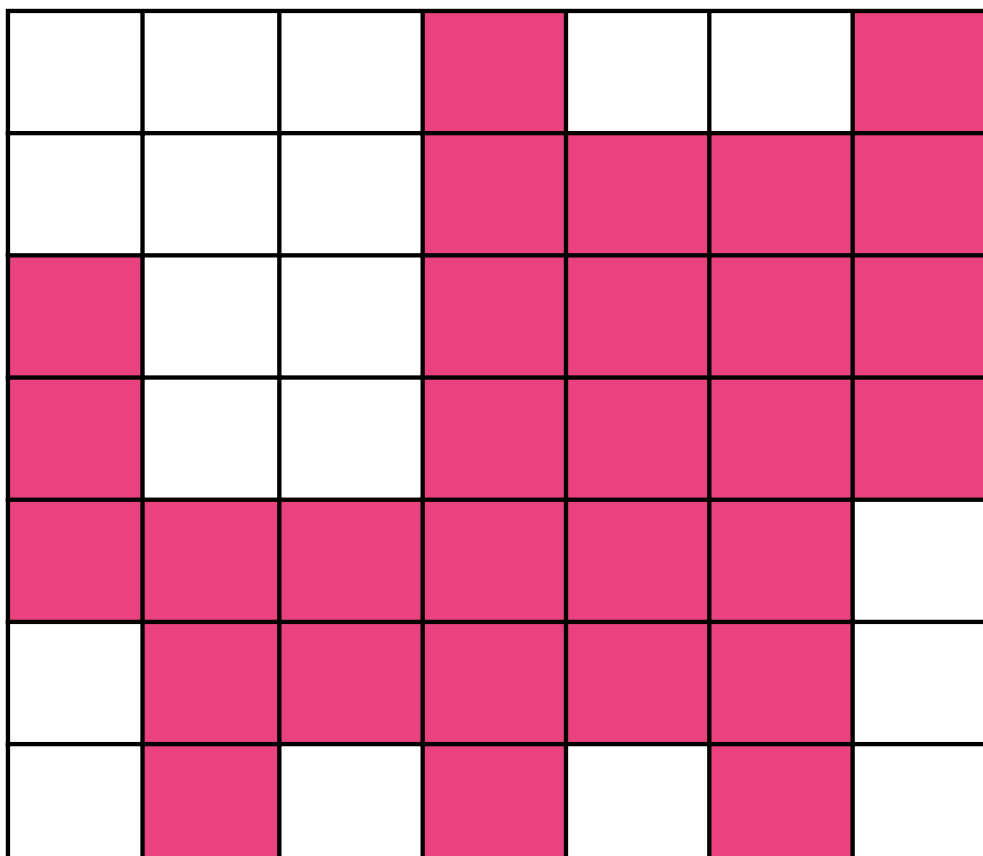
7						
1	2	1	2	1		
7						
7						
1	5	1				
2	3	2				
3	1	3				

4



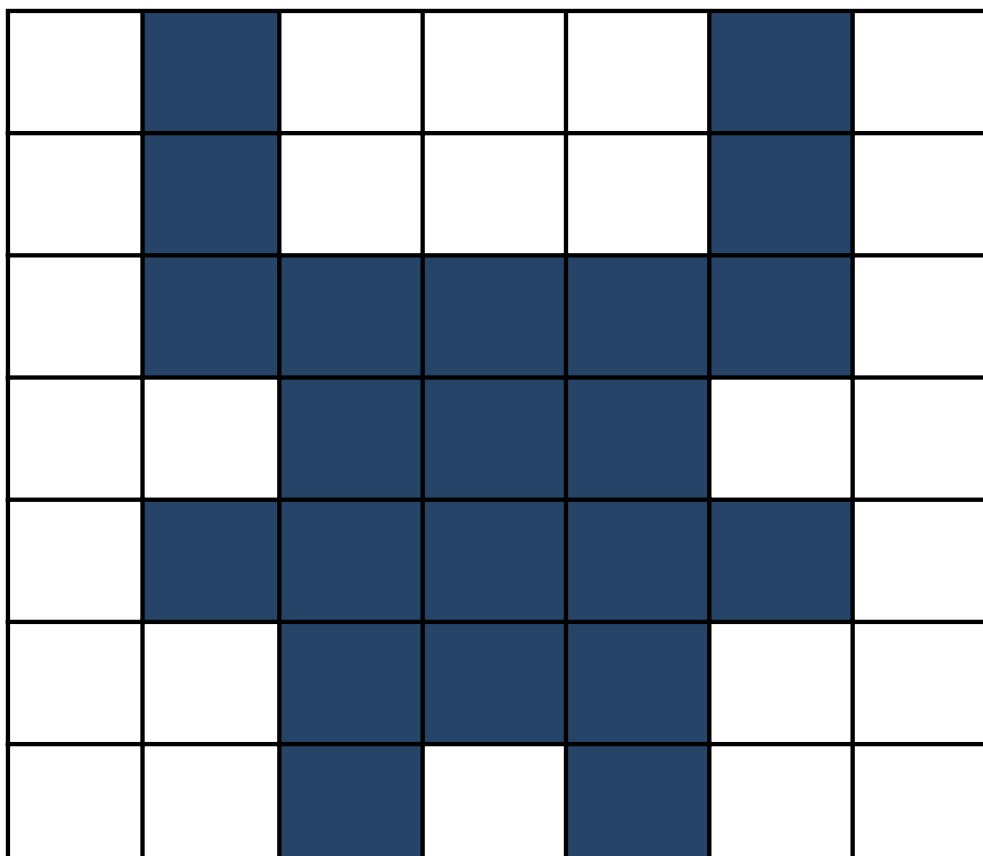
3	1	3				
2	3	2				
2	3	2				
1	5	1				
1	5	1				
1	1	1	1	1	1	1
7						

3



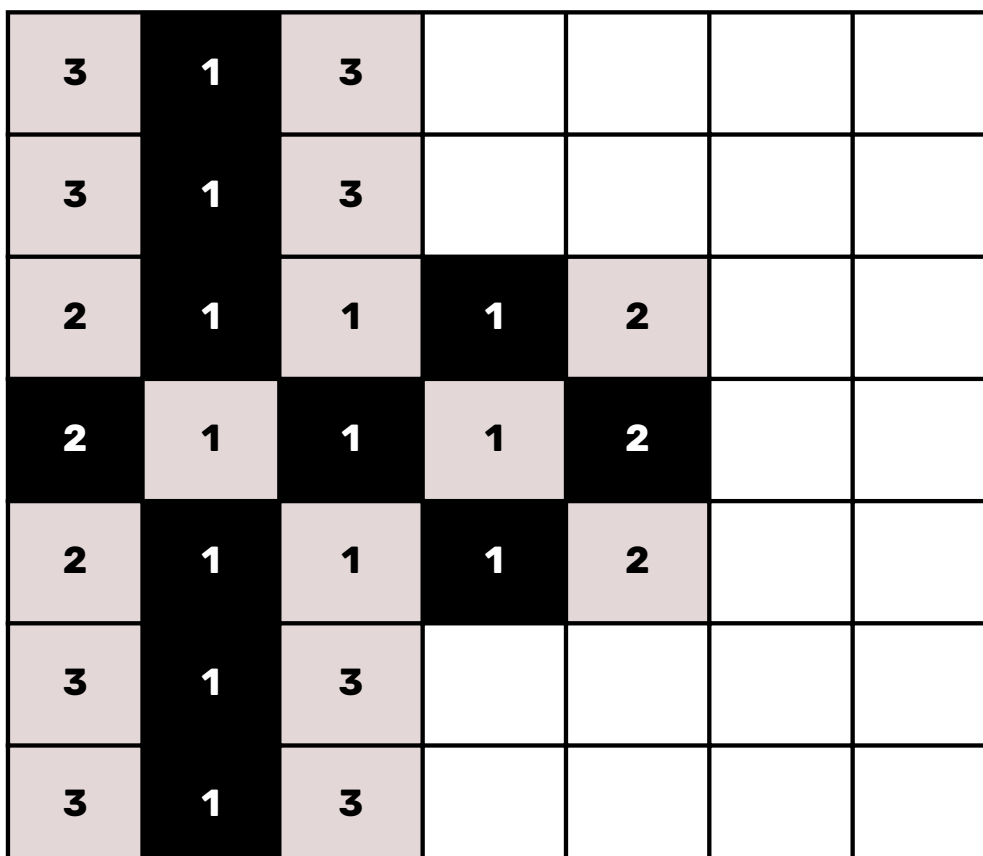
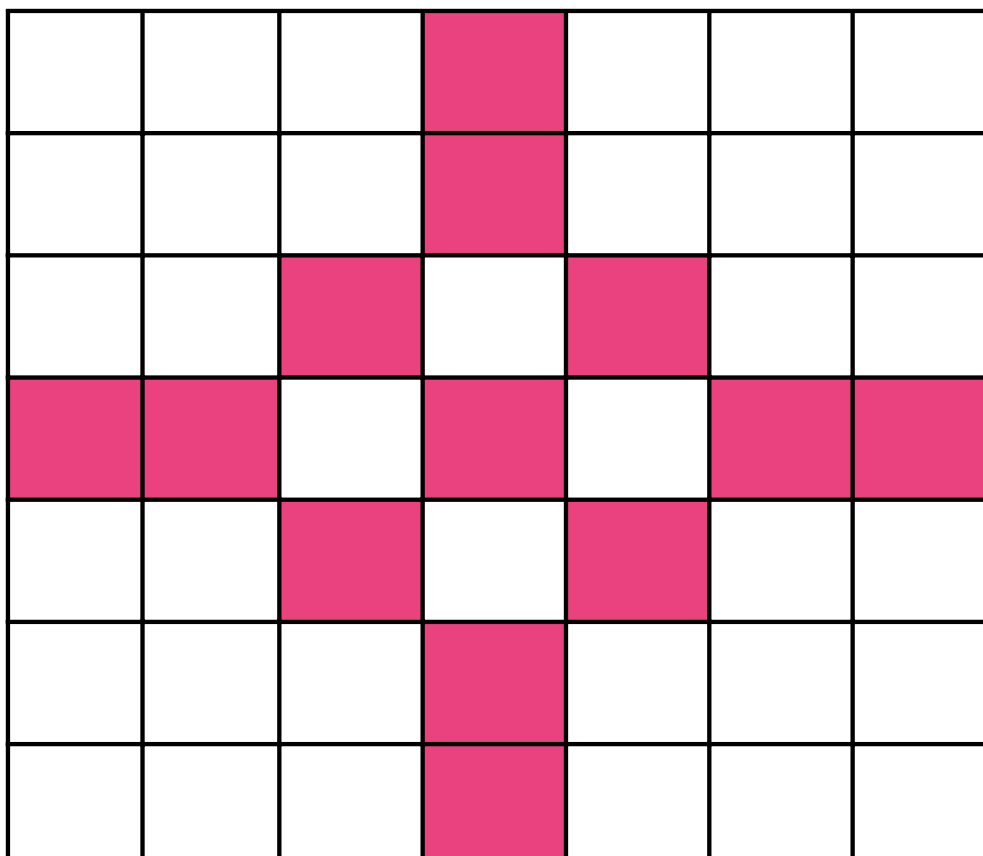
3	1	2	1			
3	4					
1	2	4				
1	2	4				
6	1					
1	5	1				
1	1	1	1	1	1	1

1

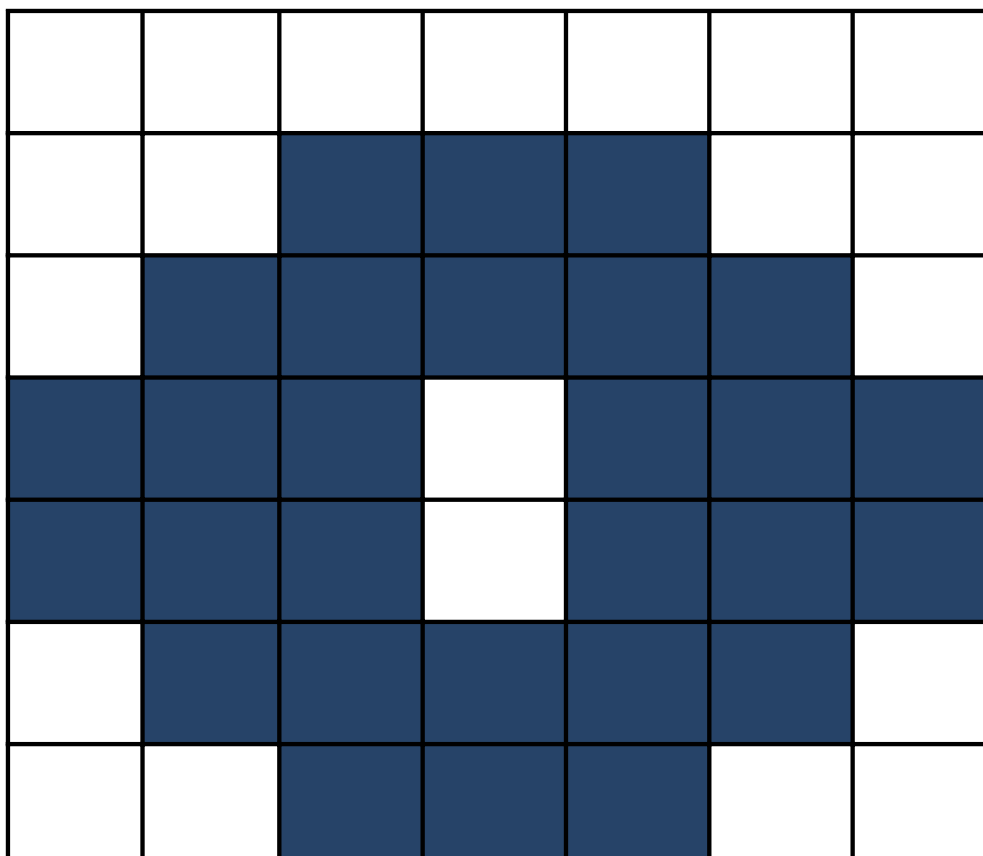


1	1	3	1	1		
1	1	3	1	1		
1	5	1				
2	3	2				
1	5	1				
2	3	2				
2	1	1	1	2		

8

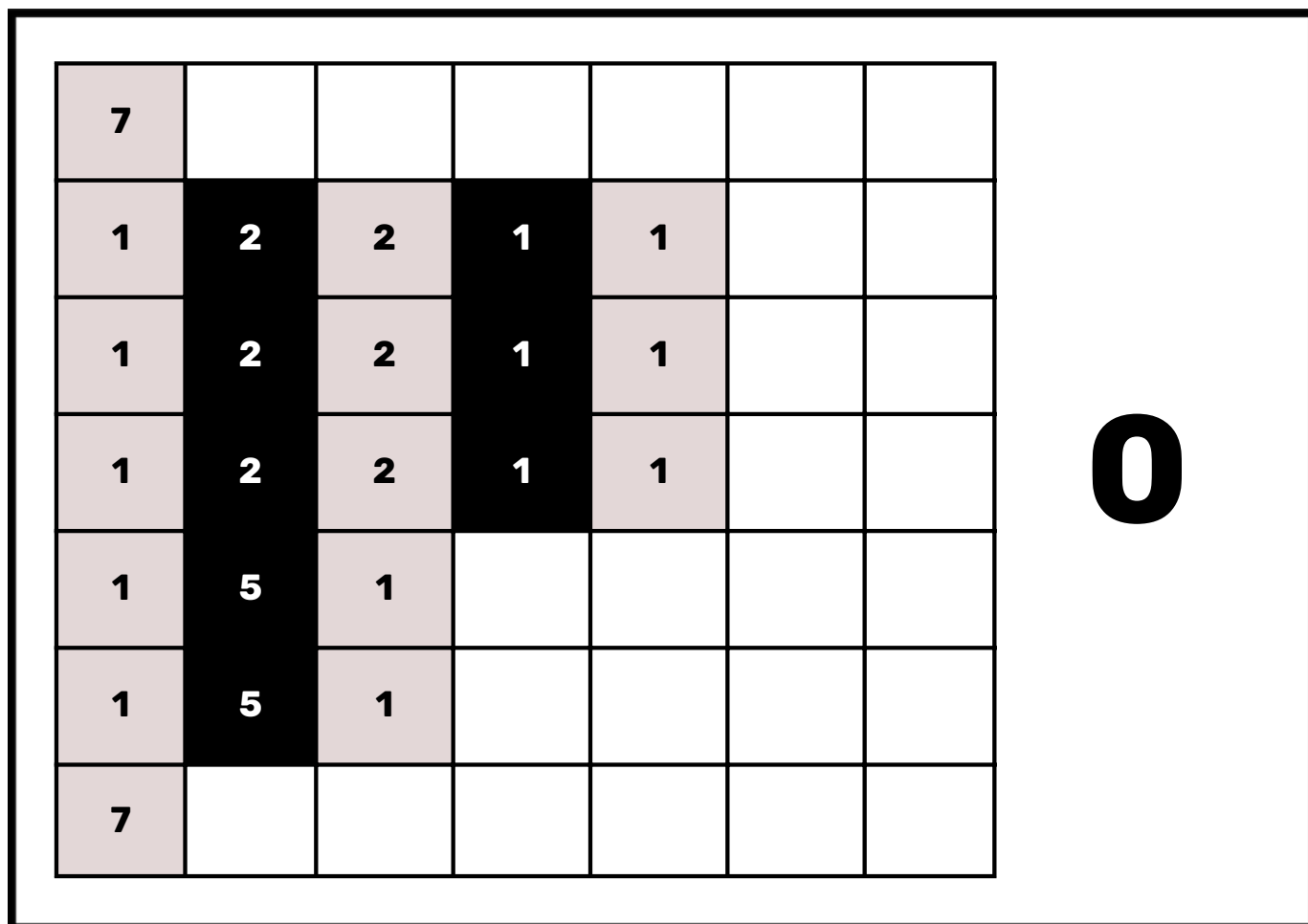
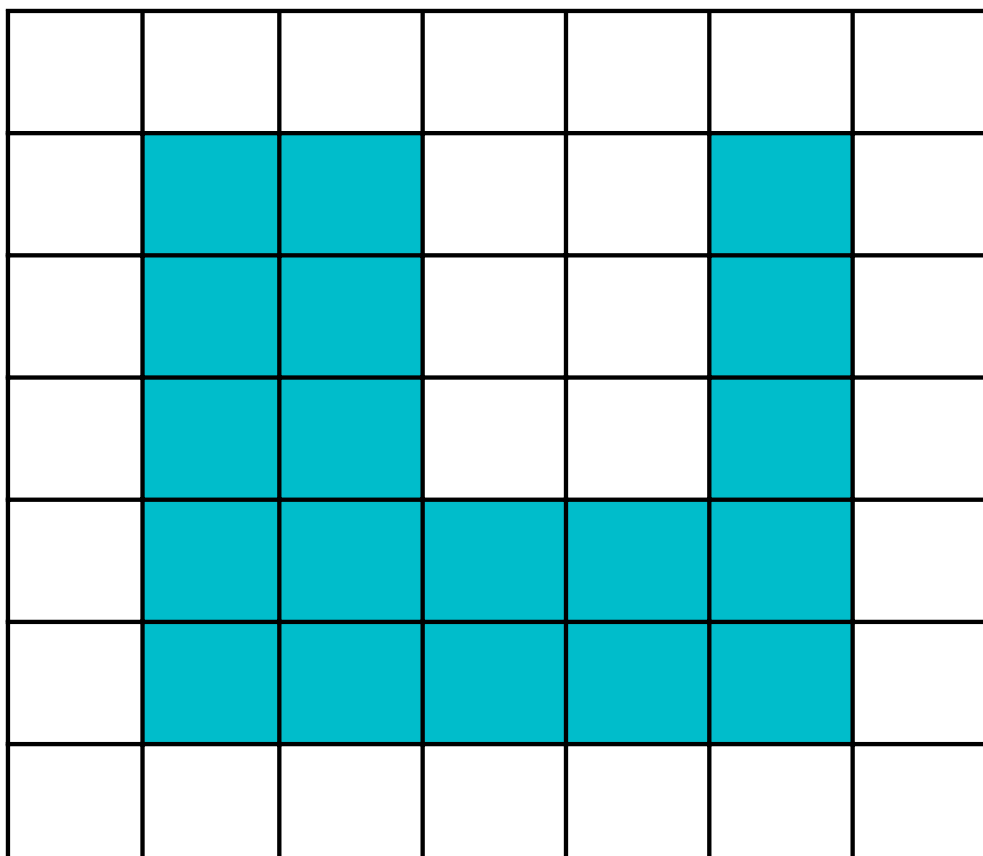


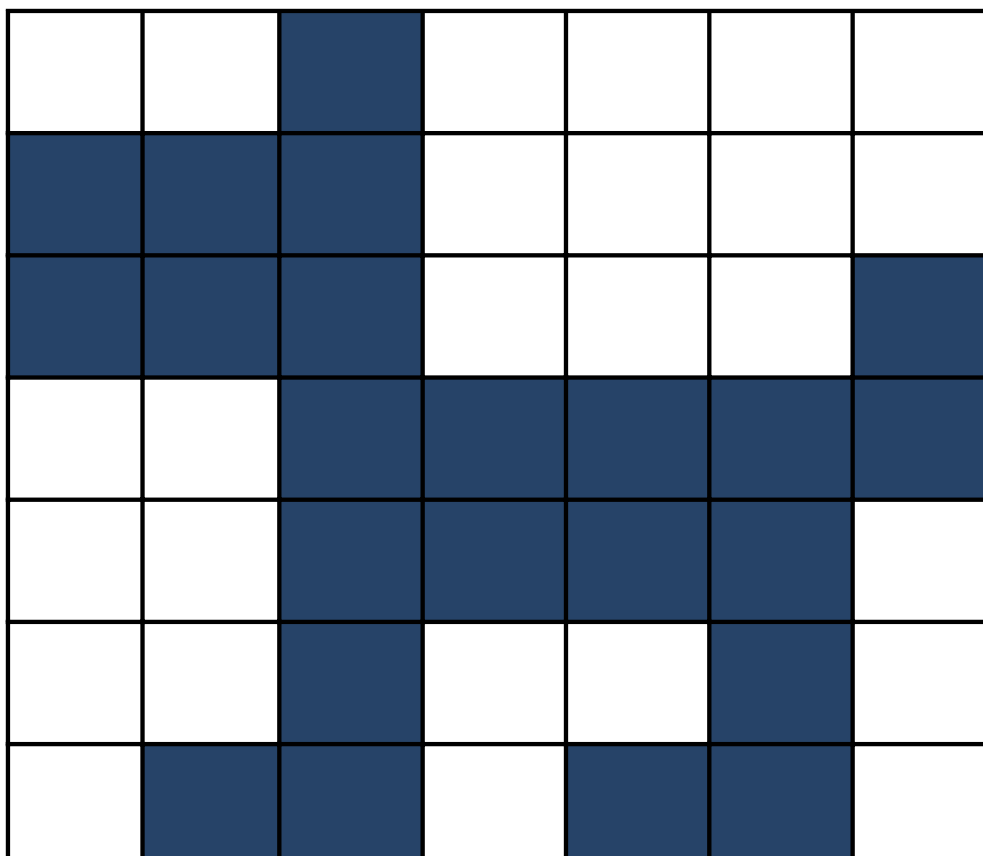
9



7						
2	3	2				
1	5	1				
3	1	3				
3	1	3				
1	5	1				
2	3	2				

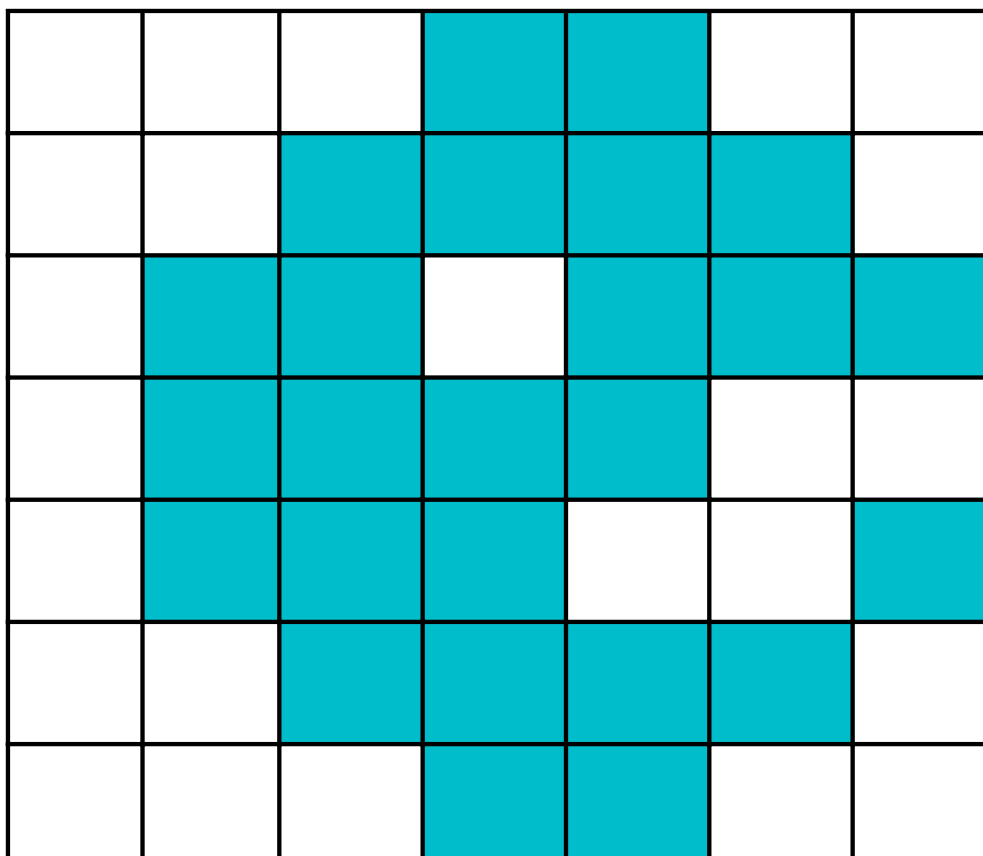
1





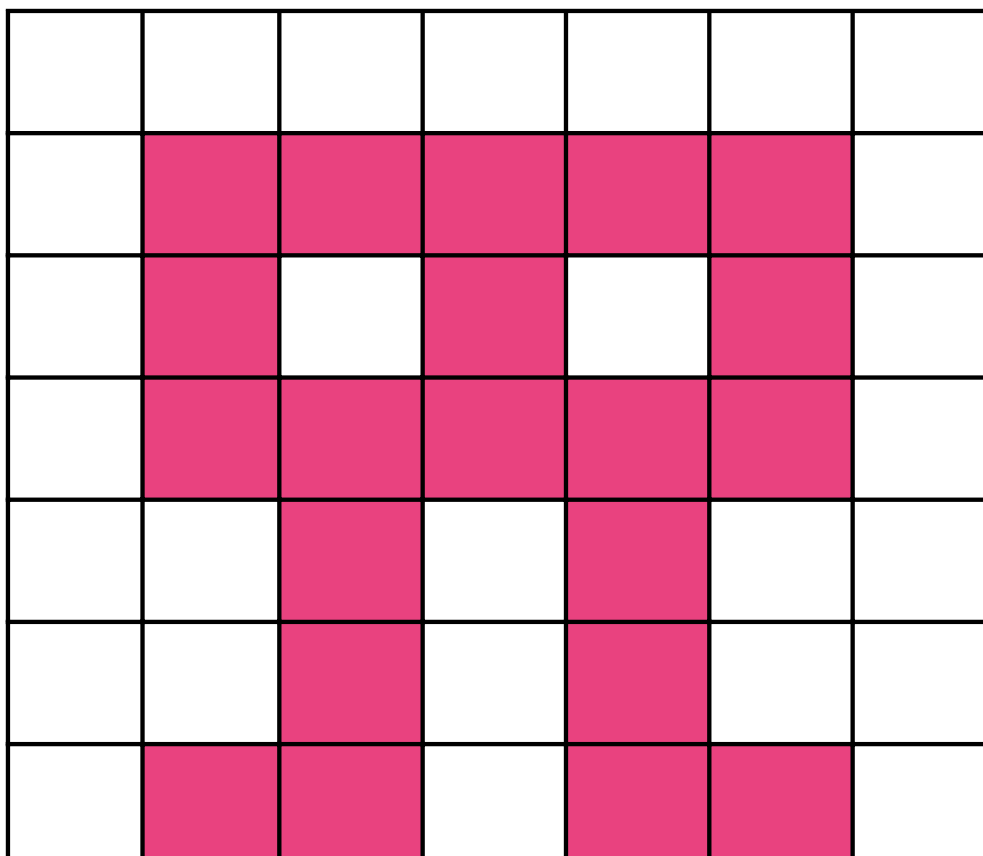
2	1	4				
3	4					
3	3	1				
2	5					
2	4	1				
2	1	1	1	1		
1	2	1	2	1		

6



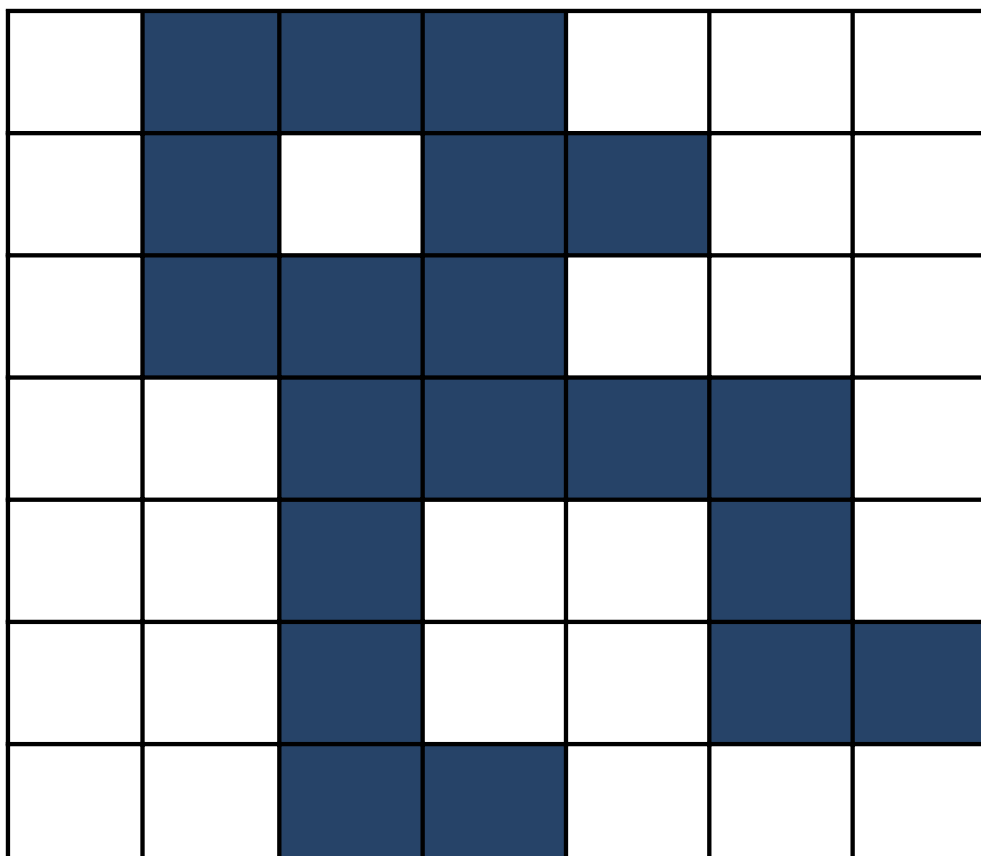
3	2	2				
2	4	1				
1	2	1	3			
1	4	2				
1	3	2	1			
2	4	1				
3	2	2				

1



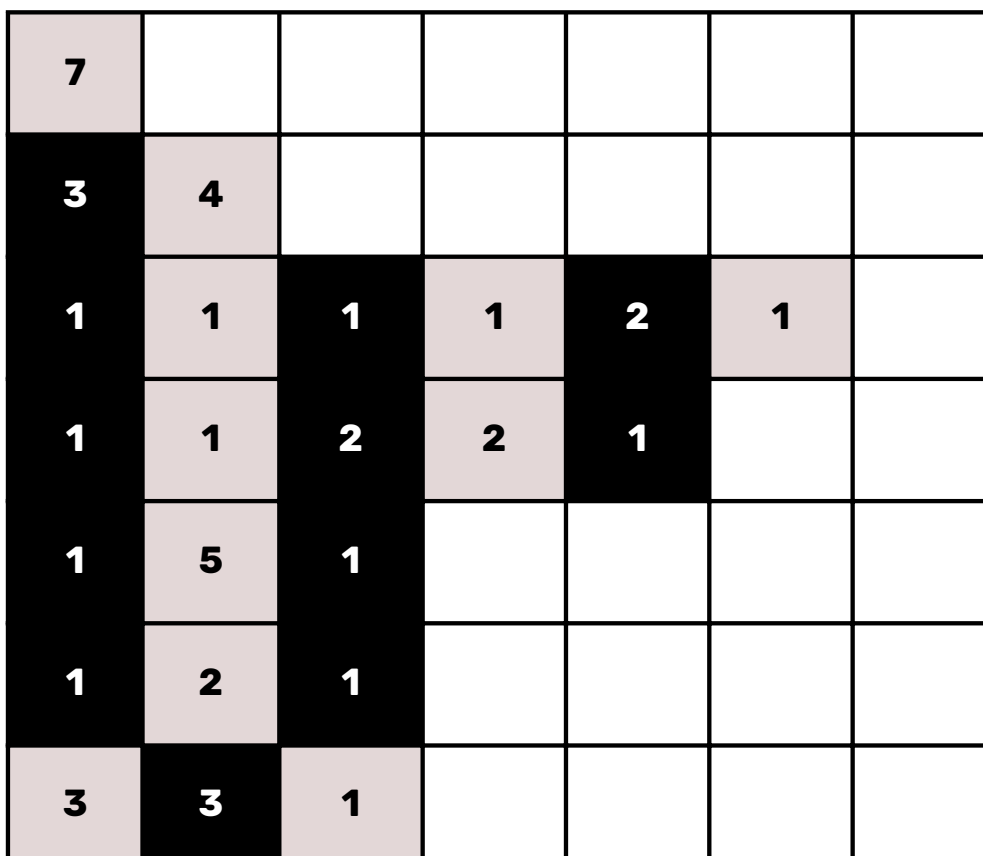
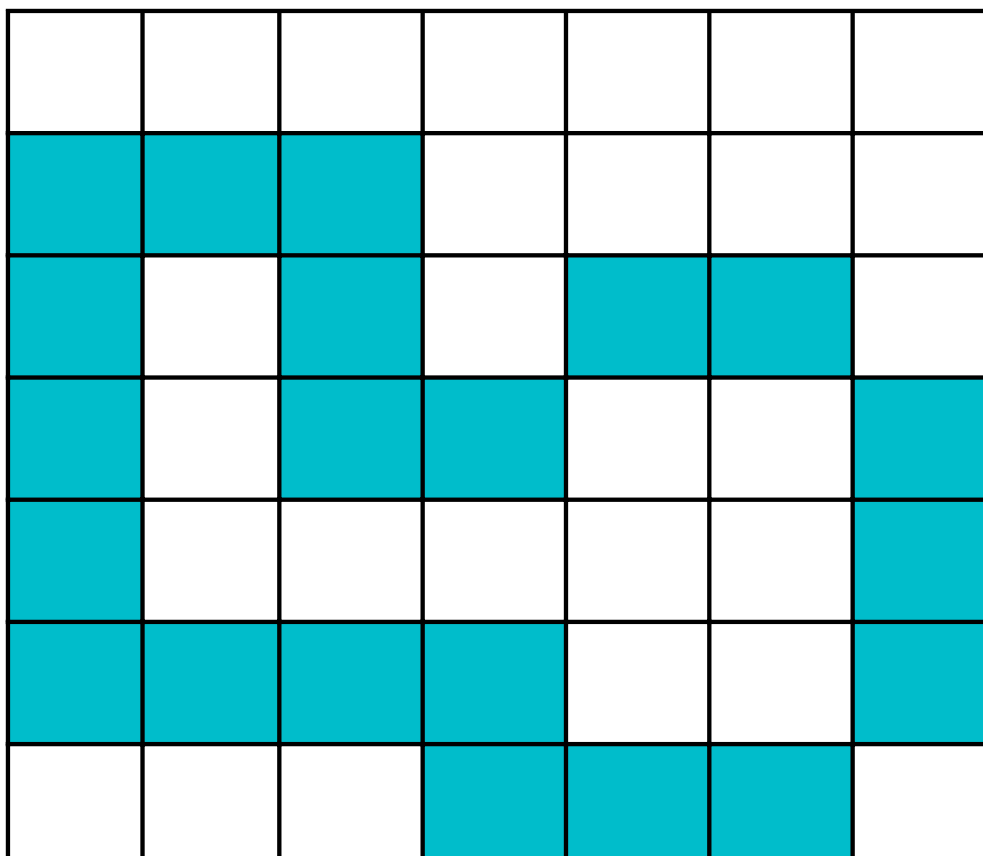
7						
1	5	1				
1	1	1	1	1	1	1
1	5	1				
2	1	1	1	2		
2	1	1	1	2		
1	2	1	2	1		

5

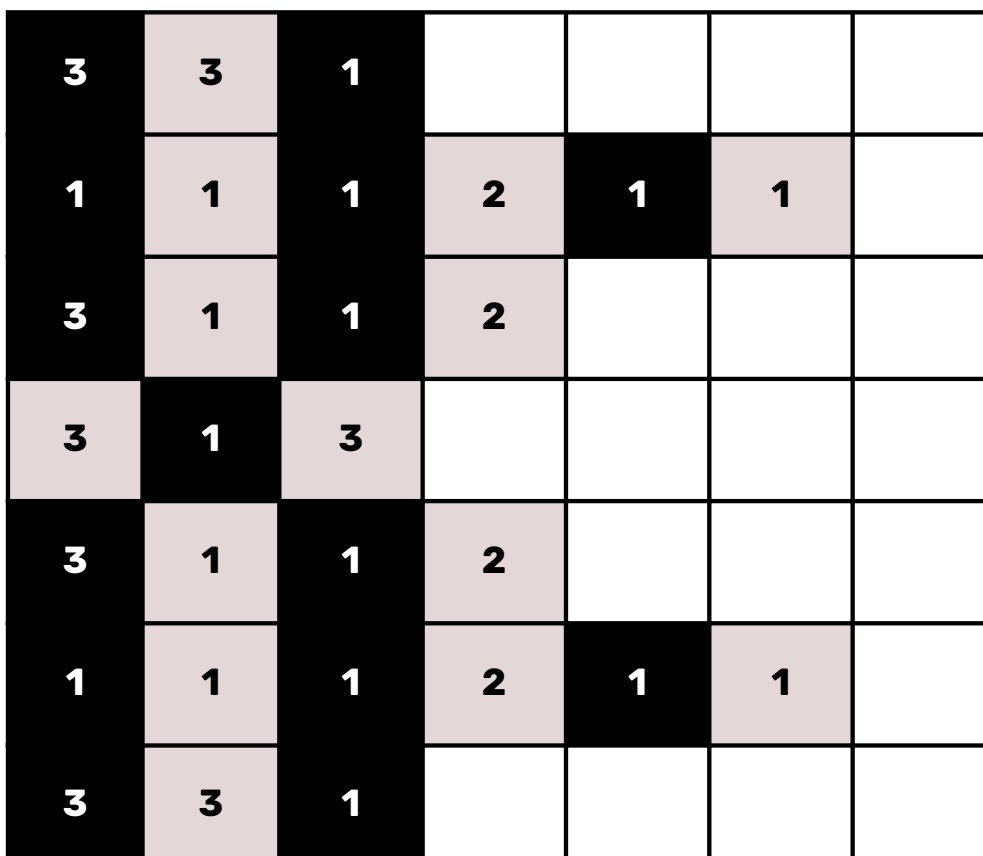
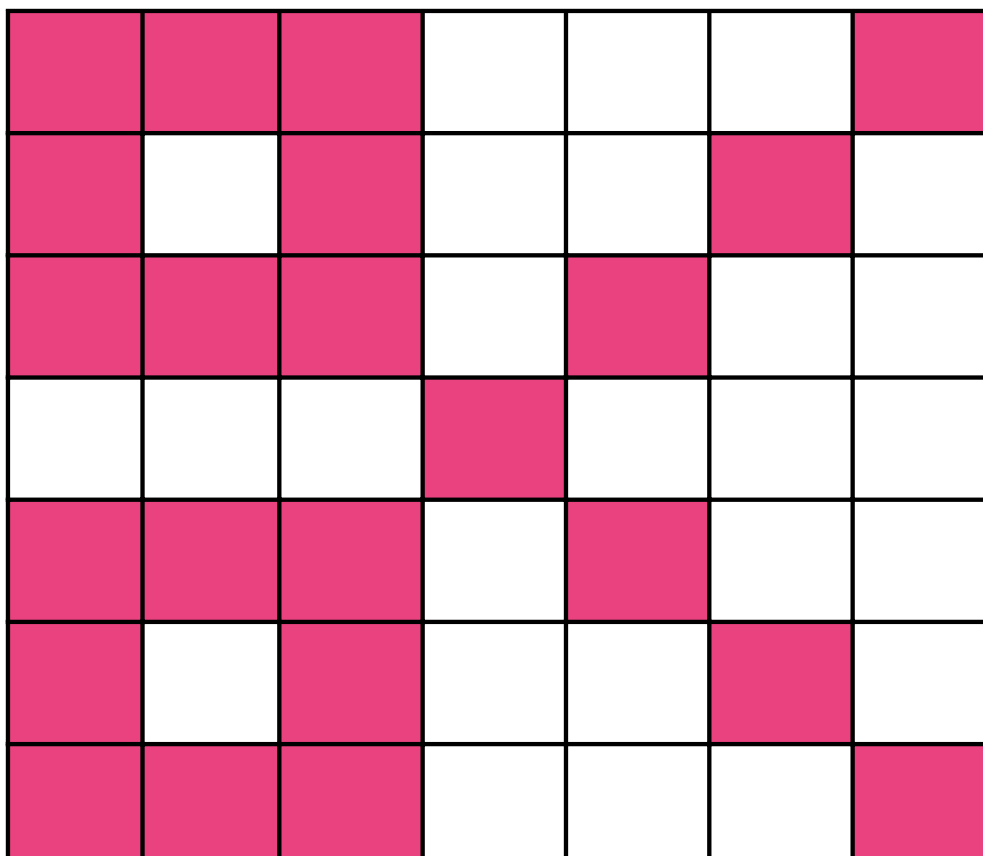


1	3	3				
1	1	1	2	2		
1	3	3				
2	4	1				
2	1	2	1	1		
2	1	2	2			
2	2	3				

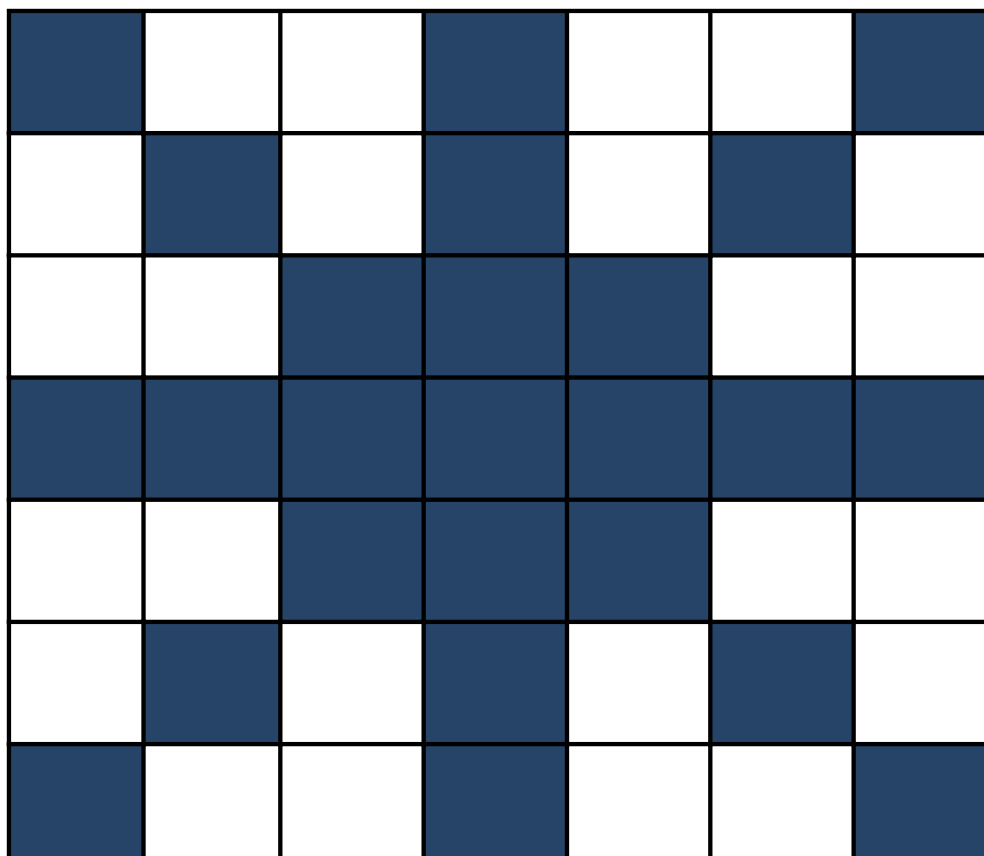
4



3

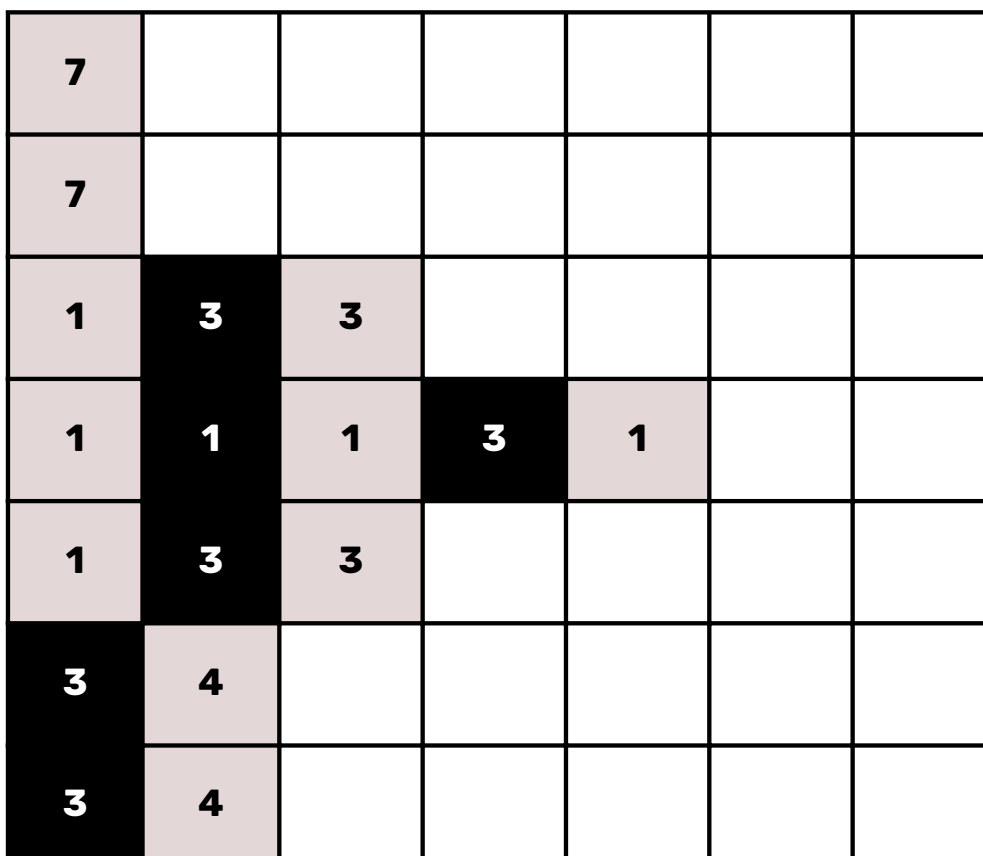
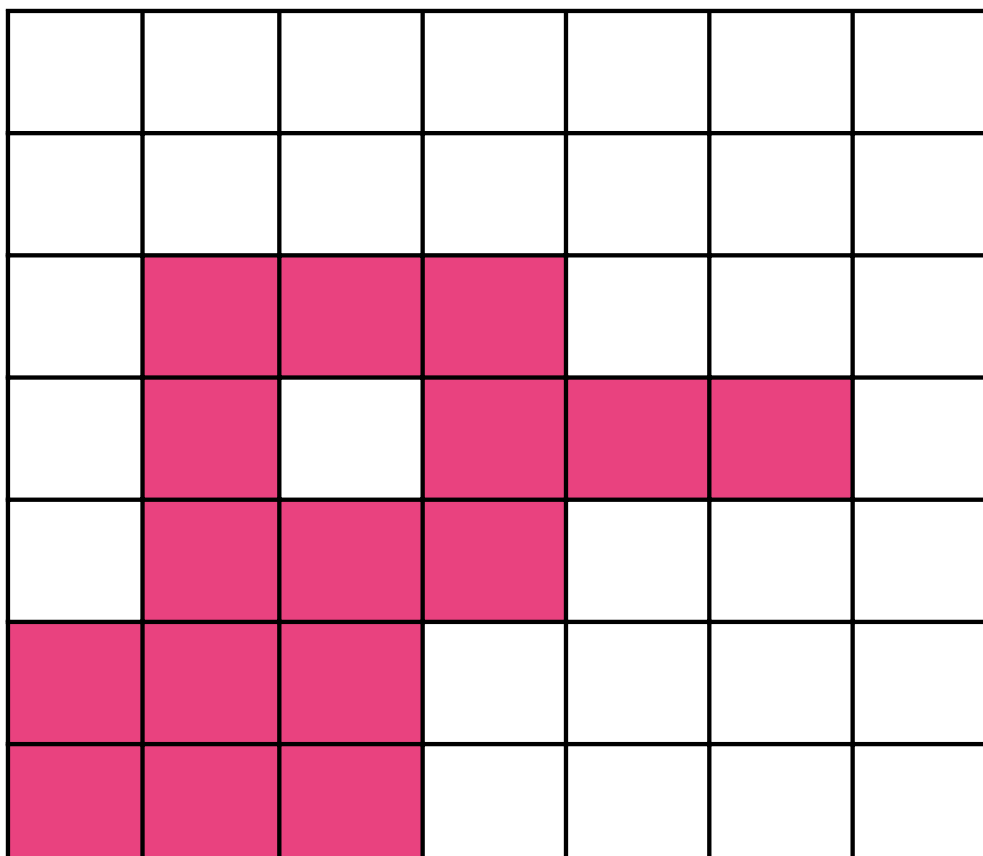


9

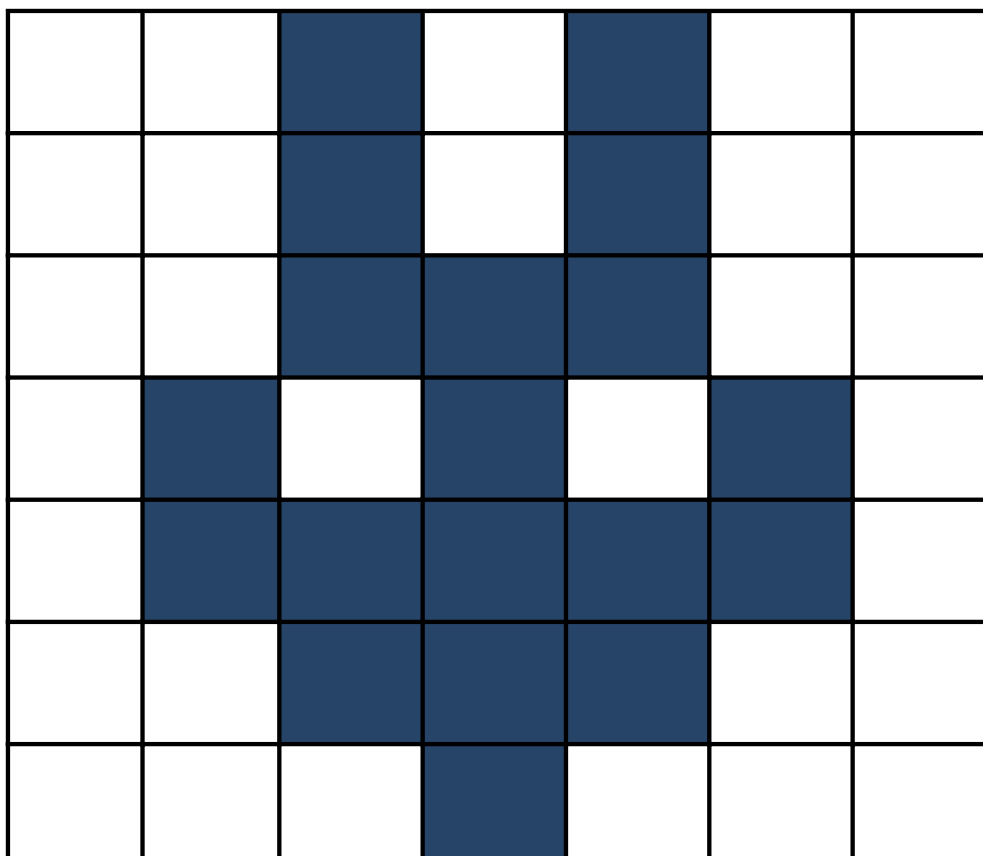


1	2	1	2	1		
1	1	1	1	1	1	1
2	3	2				
7						
2	3	2				
1	1	1	1	1	1	1
1	2	1	2	1		

8

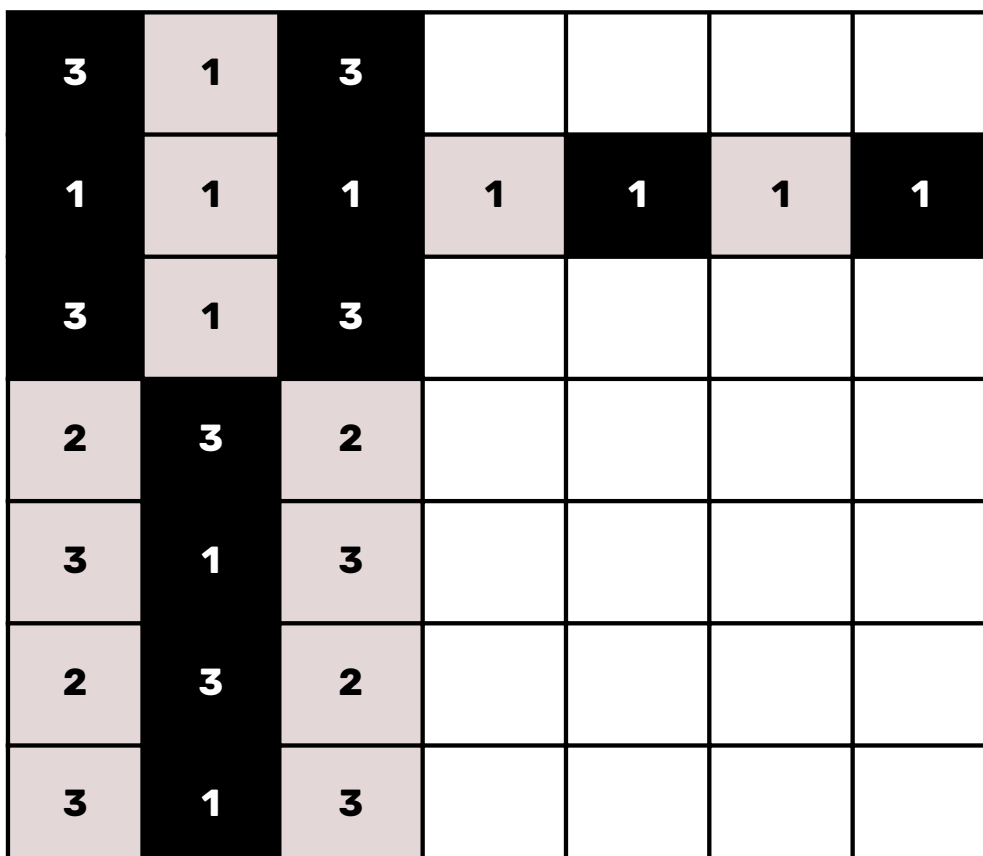
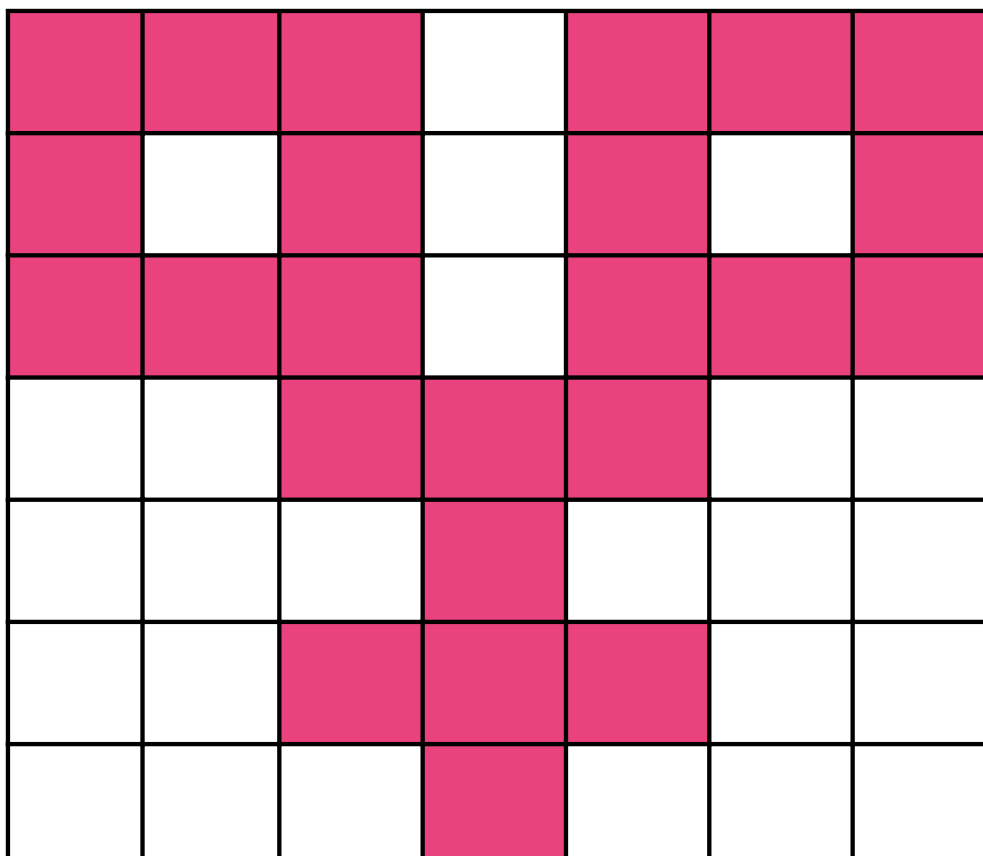


8

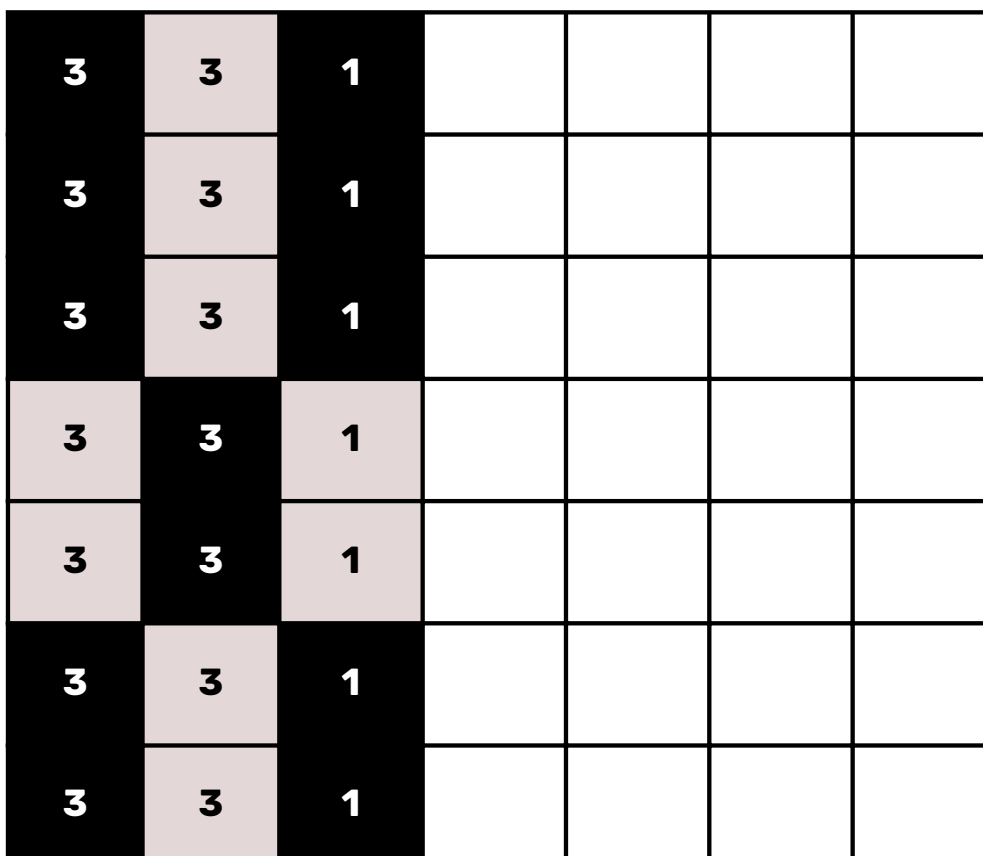
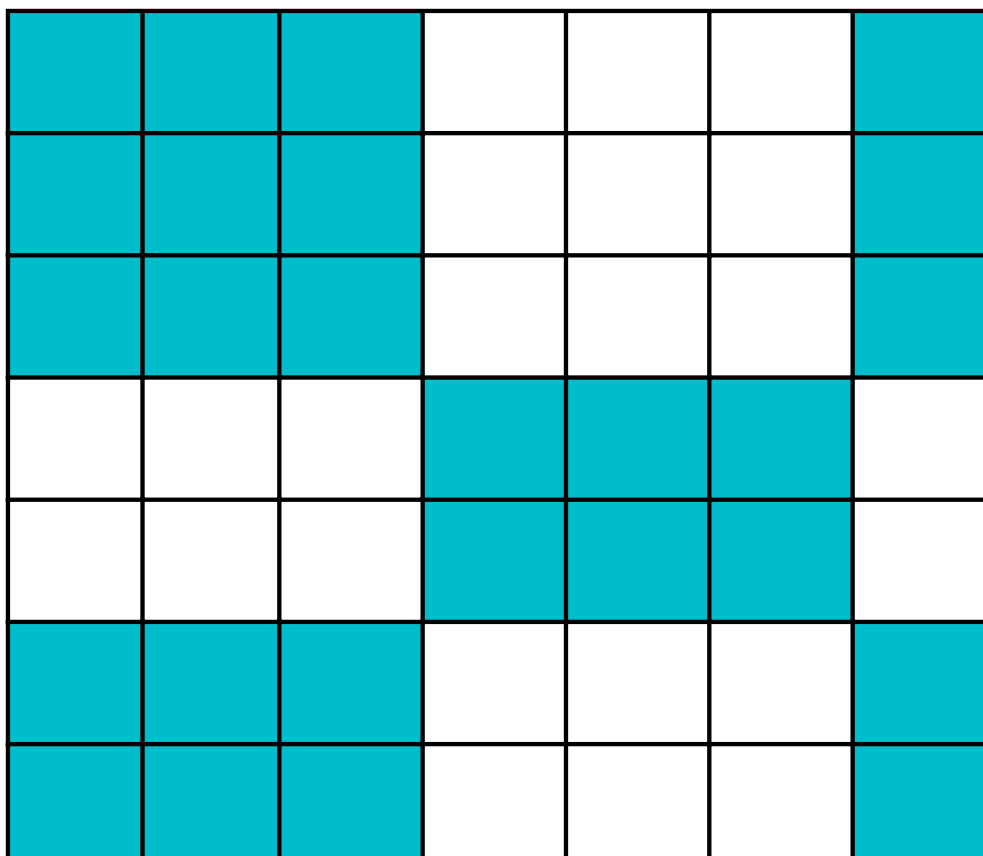


2	1	1	1	2		
2	1	1	1	2		
2	3	2				
1	1	1	1	1	1	1
1	5	1				
2	3	2				
3	1	3				

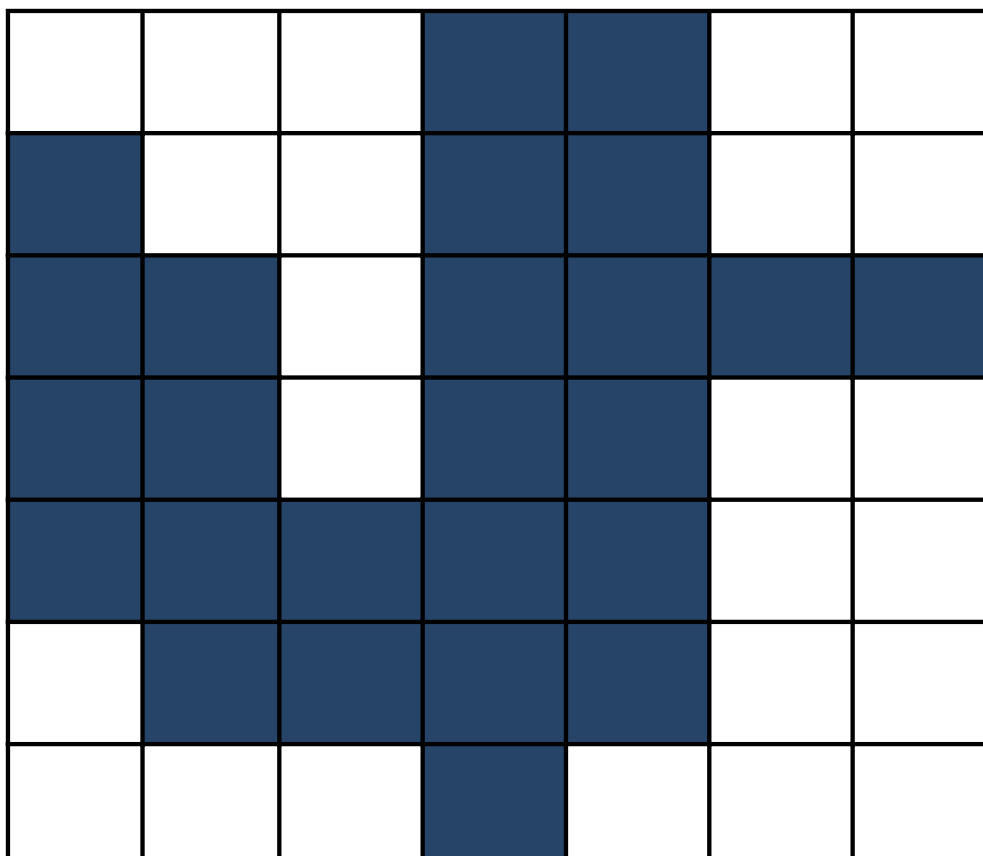
4



1

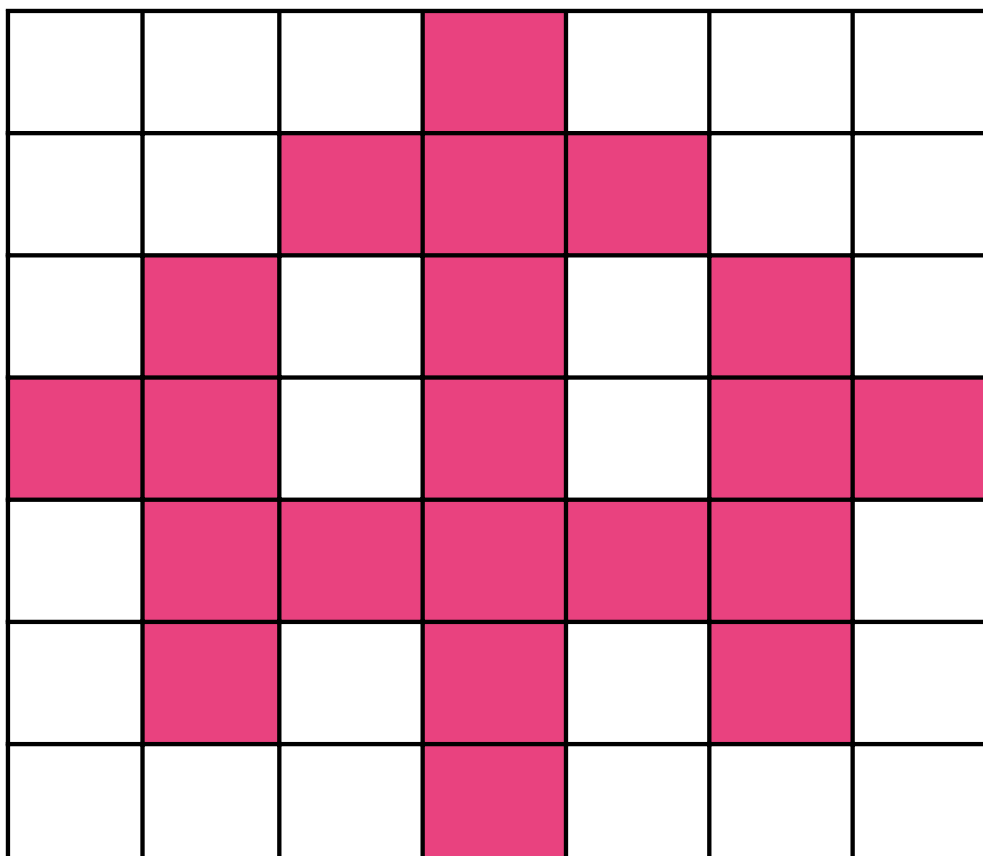


2



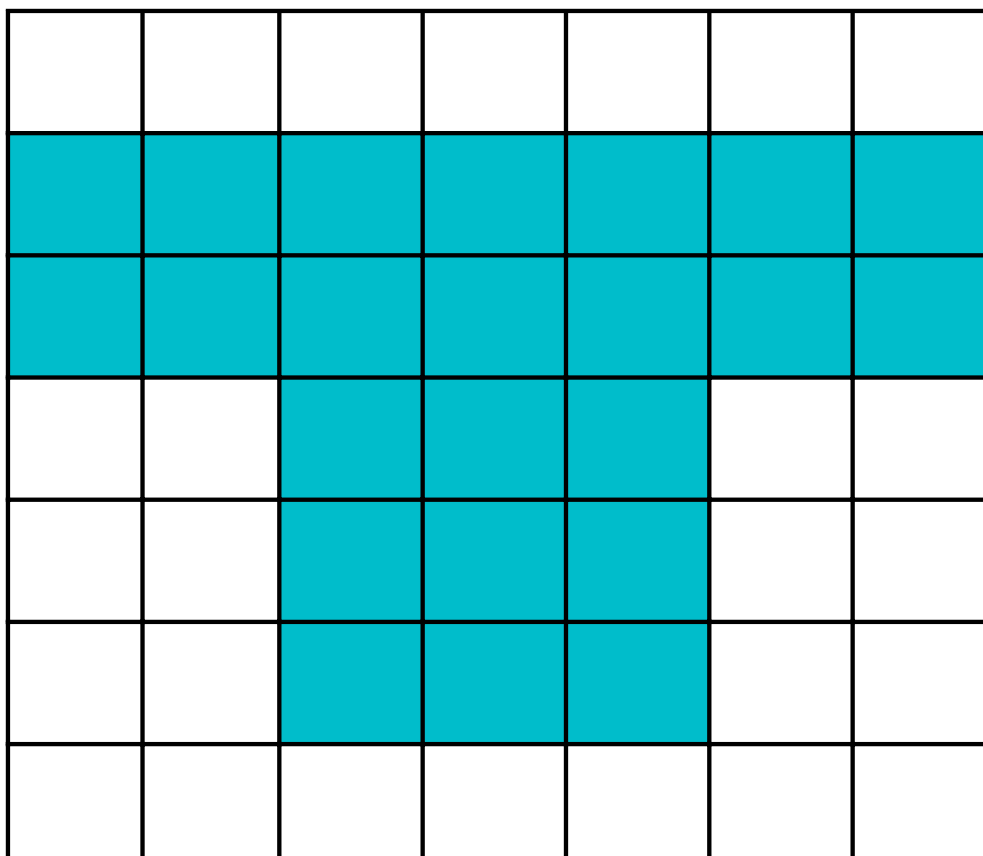
3	2	2				
1	2	2	2			
2	1	4				
2	1	2	2			
5	2					
1	4	2				
3	1	3				

7



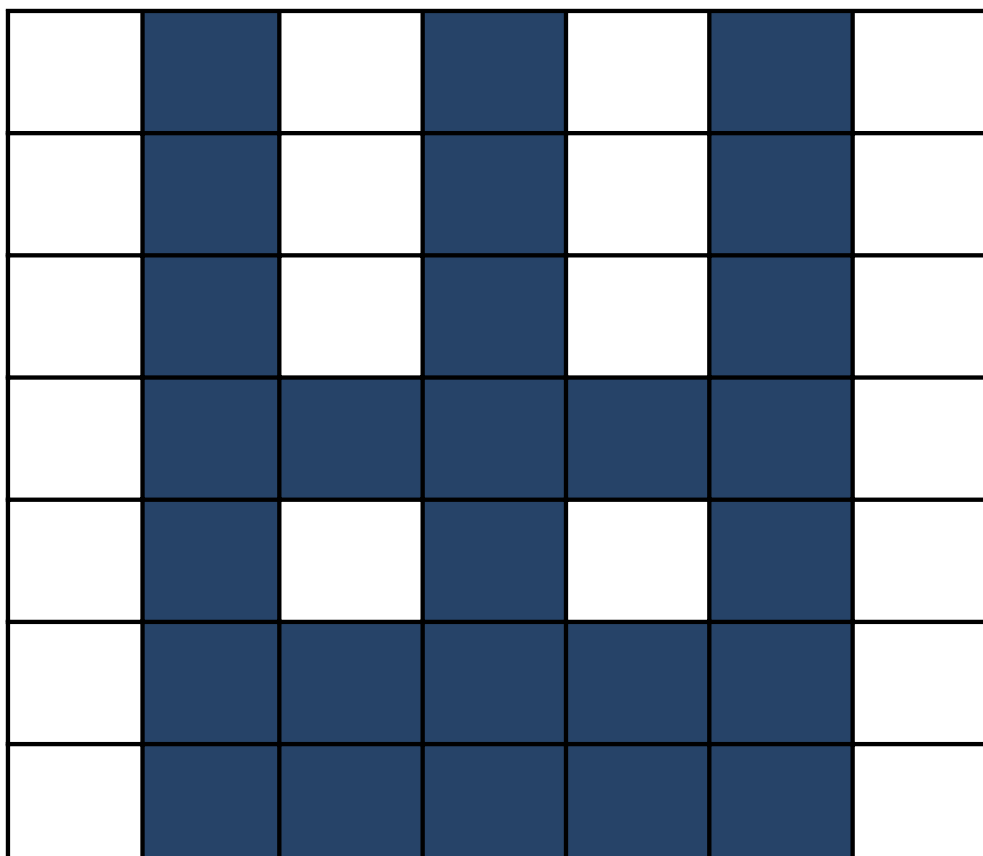
3	1	3				
2	3	2				
1	1	1	1	1	1	1
2	1	1	1	2		
1	5	1				
1	1	1	1	1	1	1
3	1	3				

1



7						
7						
7						
2	3	2				
2	3	2				
2	3	2				
7						

3



1	1	1	1	1	1	1
1	1	1	1	1	1	1
1	1	1	1	1	1	1
1	5	1				
1	1	1	1	1	1	1
1	5	1				
1	5	1				

3

